Dandelin spheres and Marden's theorem, formalised in Lean 4

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Abstract

We formalise two classical theorems about conics in Lean 4 with Mathlib. The first is Dandelin's theorem. A plane that misses the apex of a right circular cone meets it in an ellipse, a parabola or a hyperbola. The spheres inscribed in the cone and tangent to the plane touch it at the foci. In three-dimensional space we prove the classical picture: among the spheres centred on the axis and touching the cone, exactly two are tangent to the plane in the ellipse and hyperbola cases and exactly one in the parabola case; the section is *exactly* the locus of constant focal sum (ellipse) or constant absolute focal difference (hyperbola, on the double cone); for non-circular sections the focus–directrix property holds with eccentricity $e = \cos\beta/\cos\alpha$, together with its converse; the section is non-empty; the three regimes are characterised geometrically; and the planes through the apex give a point, a line or two lines. The focal identities and their converses hold in an arbitrary real inner product space; non-emptiness, the regime characterisations and the two-line apex section are proved under explicit finite-dimension assumptions. The second is Marden's theorem. For a non-degenerate triangle $a, b, c \in \mathbb{C}$, the ellipse whose foci are the critical points of $(z-a)(z-b)(z-c)$ passes through the midpoints of the sides, touches each side only there, and has the centroid as its centre. We also prove that it is the only ellipse, given by foci and focal sum, with this property. Tangency is stated metrically, and we explain why this is the right notion. The version-1 developments have about 2170 and 760 lines, respectively; version 2 adds 2287 Dandelin lines and 879 Marden lines in the Layer-14 and Layer-15 files. They contain no `sorry`, and their main theorems depend only on Lean's standard axioms. We found no earlier complete formalisation of either theorem in the sources we searched; Section 6 records the scope of that search and one unresolved lead. Version 2 restates the main theorems with Mathlib-only types, each with an explicit non-degenerate witness, and compares the development with textbook statements formalised independently; the comparison adds further proved statements and finds no false one.

1 Introduction

We work in Lean 4 [12] with Mathlib [10]. Pascal's hexagon theorem is treated in a companion paper [1]. This paper deals with two other classical results about conics. Both are on the "1000+ theorems" list that Mathlib maintains, and at the Mathlib version we use neither entry has a formal declaration attached [9].

Dandelin's theorem (1822) [2] links the definition of a conic as a plane section of a cone to its definition by foci. Suppose a plane cuts one nappe of a right circular cone in a closed curve. There are two spheres that are inscribed in the cone and tangent to the plane, one on each side of it. If F_1,

F_2 are the points of tangency, then every point X of the curve satisfies $|XF_1|+|XF_2| = \text{const}$. The proof uses only the fact that all tangent segments from a point to a sphere have the same length. The same construction gives the hyperbola, with one sphere in each nappe, and the parabola, with one sphere. The circle along which a sphere touches the cone lies in a plane σ; for a non-circular section, σ meets the cutting plane in the directrix. For textbook accounts see Hilbert and Cohn-Vossen [4].

Marden's theorem [7, 8] is named after Marden, and it goes back to Siebeck [13]. Kalman gives an elementary proof [6], and Minda and Phelps give another [11]. It states the following. Let $a, b, c \in \mathbb{C}$ be non-collinear and let $p(z) = (z-a)(z-b)(z-c)$. Then the roots of p' are the foci of an ellipse that is tangent to the three sides of the triangle at their midpoints. This ellipse is the *Steiner inellipse*, and its centre is the centroid.

Contributions.

- A Lean formalisation of Dandelin's theorem for all three non-degenerate conics (Section 3). The focal identities hold in any real inner product space. The hypotheses concern only the cone and the plane. The spheres, the foci and (for non-circular sections) the directrices are constructed, and we prove that the tangent spheres are unique within the family of axis spheres touching the cone (in $\mathbb{R}^3$, a Layer-15 lemma extends this to every sphere that meets each generator in exactly one point; Section 7.1). Every characterisation is proved in both directions: the section *equals* the focal locus.
- The regime classification: the bounded-section inequality is equivalent to the section lying in one nappe and being bounded, and the hyperbola inequality is equivalent to meeting both nappes. We also prove non-emptiness, and treat the degenerate sections by planes through the apex. These results assume that E is finite-dimensional with the stated lower bound on its dimension, except where noted.
- A Lean formalisation of Marden's theorem with a metric notion of tangency, together with the uniqueness of the inellipse among all ellipses given by two foci and a focal sum (Section 4).
- Lemmas that help a reader check that the formal statements mean what they say (Section 5). They include a Lean counterexample showing that one of our statements cannot be strengthened in the obvious way.
- (Version 2.) Mathlib-only restatements of the main theorems with witnesses (Layer 14), and a comparison with independently formalised textbook statements (Layer 15), Section 7.1.

Availability. The code is at `https://github.com/jpbald93/dandelin` and `https://github.com/jpbald93/marden`. Line references are to commits `205e480` and `bd2cd3c` respectively. Version 2 adds the Layer-14 and Layer-15 files (`Dandelin/L14.lean`, `Dandelin/L15Classical.lean`, `Marden/L14.lean`, `Marden/L15Classical.lean`) in a new commit of each repository (Dandelin `89309188e77f1564fd31cc9a2d245dbba1b34f8c`, Marden `b4fbe75198e728699764a9b44e8f9981ba589299`). That commit also adds two import lines to each root file and updates `gate.sh` and the README. The earlier Lean files and statements are unchanged, so the line references above remain valid. The toolchain is Lean `v4.33.1`. Both projects pin Mathlib at tag `v4.33.1` (revision `0df444a`) in `lake-manifest.json`. Supporting material (the Layer-14 and Layer-15 reports, the blind writer's statements and report, the transcript audit, and the reviews of this version with the fixes made in response) is in the folder `Conics-dandelin-marden` of the repository `https://github.com/jpbald93/LEAN-verification`.

Table 1: The main formal statements. Namespaces `Dandelin` (upper part) and `Marden` (lower part). "Thm" refers to the numbered results of this paper.

Lean name	In this paper	Source
`tangent_lengths_eq`	equal tangents, §3.1	`Basic.lean:29`
`cone_tangent_sq`	Lemma 1	`Basic.lean:53`
`cone_outside_sphere`	Lemma 19	`Basic.lean:63`
`tangencyPoint_mem_sphere`	Lemma 19	`Basic.lean:85`
`tangencyPoint_perp`	Lemma 19	`Basic.lean:95`
`dandelin_of_between_isTangentAt`	Remark 21	`Basic.lean:146`
`dandelin`	Theorem 2(iv), $\Rightarrow$, normalised	`Basic.lean:216`
`tangent_iff`	Theorem 2(i), normalised ($s < m$, $c > 0$)	`Existence.lean:64`
`dandelin_converse`	Theorem 2(iv), $\Leftarrow$	`Existence.lean:155`
`dandelin_exists`	Theorem 2	`Existence.lean:230`
`section_bounded`	Theorem 9	`Existence.lean:277`
`infDist_directrix`	Lemma 5	`Directrix.lean:72`
`focal_dist_eq_ecc_mul`	Theorem 6	`Directrix.lean:148`
`ecc_lt_one_iff`	Theorem 6, $e < 1$	`Directrix.lean:160`
`ecc_eq_one`	Theorem 6, $e = 1$	`Directrix.lean:168`
`one_lt_ecc_iff`	Theorem 6, $e > 1$	`Directrix.lean:174`
`tangent_of_proj`	Lemma 20	`Directrix.lean:192`
`directrix_ellipse`	Theorem 6	`Directrix.lean:206`
`directrix_parabola`	Theorem 7	`Directrix.lean:247`
`mem_cone2_iff`	§2	`Hyperbola.lean:25`
`hyp_upper`	Theorem 4(ii), upper	`Hyperbola.lean:136`
`hyp_lower`	Theorem 4(ii), lower	`Hyperbola.lean:157`
`hyp_converse`	Theorem 4(iii), $\Leftarrow$	`Hyperbola.lean:192`
`dandelin_hyperbola`	Theorem 4	`Hyperbola.lean:283`
`hyp_both_nappes`	Theorem 9(c)	`Regime.lean:222`
`ellipse_nonempty`	Theorem 9(d)	`Regime.lean:255`
`section_unbounded`	Theorem 9(b)	`Regime.lean:330`
`one_nappe`	Theorem 9(a)	`Regime.lean:373`
`bounded_iff`	Theorem 9(b)	`Regime.lean:407`
`ellipse_regime_iff`	Theorem 9(b)	`Regime.lean:426`
`hyperbola_regime_iff`	Theorem 9(c)	`Regime.lean:442`
`parabola_regime`	Theorem 9(e)	`Regime.lean:451`
`regime_R3`	Theorem 9 in $\mathbb{R}^3$	`Regime.lean:475`
`directrix_converse`	Theorem 8	`Degenerate.lean:30`
`directrix_iff_ellipse`	Theorem 8(a)	`Degenerate.lean:88`
`directrix_iff_parabola`	Theorem 8(a)	`Degenerate.lean:113`
`directrix_iff_hyperbola`	Theorem 8(b)	`Degenerate.lean:133`
`apex_section_point`	Theorem 10(a)	`Degenerate.lean:226`
`apex_section_line`	Theorem 10(b)	`Degenerate.lean:252`
`apex_section_two_lines`	Theorem 10(c)	`Degenerate.lean:369`

Lean name	In this paper	Source
`apex_two_lines_subset`	Theorem 10, remark	`Degenerate.lean:443`
`apex_section_R3`	Theorem 10 in $\mathbb{R}^3$	`Degenerate.lean:470`
`midpoint_mul`	Lemma 12	`Basic.lean:24`
`focalSum_midpoint_sq`	Lemma 14	`Basic.lean:37`
`focalSum_midpoints_eq`	Theorem 11(i)	`Basic.lean:59`
`focal_le_of_mul`	Lemma 13	`Tangent.lean:73`
`focal_lt_of_mul`	Lemma 13, strict	`Tangent.lean:100`
`vieta`	§4.1	`Main.lean:31`
`exists_critical_points`	Lemma 22	`Main.lean:44`
`marden`	Theorem 11	`Main.lean:136`
`reflection_of_min`	Lemma 17	`Unique.lean:74`
`foci_of_reflection`	Lemma 18	`Unique.lean:194`
`steiner_inellipse_unique`	Theorem 15, weak form	`Unique.lean:233`
`nondeg_of_midpoints`	Lemma 22	`Unique.lean:322`
`steiner_unique`	Theorem 15	`Unique.lean:354`
`steiner_inellipse_iff`	Corollary 16	`Unique.lean:373`

2 Conventions

Ambient space. Throughout the Dandelin development, E is a real inner product space (`[NormedAddCommGroup E] [InnerProductSpace ℝ E]`), and points are vectors of E. There is no finite-dimensionality assumption unless one is stated. Where one is stated, it is `[FiniteDimensional ℝ E]` together with a bound on `Module.finrank ℝ E`, and we write "E finite-dimensional, $\dim E \geq k$". The intended case is `EuclideanSpace ℝ (Fin 3)`, and results that need a dimension bound are also specialised to it.

Cone. A cone is given by an apex V, a unit axis vector u, and $k = \cos\alpha$, $s = \sin\alpha$, where α is the half-angle. The hypotheses are $k^2 + s^2 = 1$, $k > 0$ and (usually) $s > 0$, so that $0 < \alpha < \pi/2$. One nappe and the double cone are

$$\texttt{cone V u k} = \{X : \langle X - V, u\rangle = k\|X - V\|\}, \qquad \texttt{cone2 V u k} = \{X : \langle X - V, u\rangle^2 = k^2\|X - V\|^2\}.$$

Both contain V. `mem_cone2_iff` proves that `cone2 V u k` = `cone V u k` ∪ `cone V (-u) k`. The axis sphere with parameter $d \in \mathbb{R}$ has centre $C_d = V + du$ and radius $|d|s$, which is the distance from C_d to every generator. For $d > 0$ it is inscribed in the upper nappe, and for $d < 0$ in the lower one (Lemma 19). All uniqueness statements about spheres in the main files are about this one-parameter family $\{(C_d, |d|s)\}$; the main files do not classify arbitrary centres and radii satisfying an independent notion of being inscribed. The Layer-15 file adds this classification in $\mathbb{R}^3$: a sphere of positive radius that meets every generator of the double cone in exactly one point is an axis sphere $(C_d, |d|s)$ with $d \neq 0$ (`Dandelin.Classical.l15_tangent_cone_iff`, Section 7.1).

Plane. `plane p₀ n` $= \{X : \langle X - p_0, n\rangle = 0\}$ with $\|n\| = 1$. We write

$$m = \langle u, n\rangle, \qquad c = \langle p_0 - V, n\rangle.$$

So $|m| = \sin\beta$, where β is the angle between the plane and the axis, and c is the signed distance from V to the plane. The plane misses the apex exactly when $c \neq 0$.

Tangency. A sphere with centre C and radius r is tangent to the plane when $|\langle C - p_0, n\rangle| = r$, that is, when the distance from the centre to the plane equals the radius. Its tangency point is then the projection $F = C - \langle C - p_0, n\rangle\, n$ of the centre onto the plane. Lemma 20 shows that this matches the pointwise definition used elsewhere: F is on the sphere, and $X - F \perp F - C$ for every X in the plane.

Distances to sets. Distances to the directrix use Mathlib's `Metric.infDist`. Because `infDist` $X\,\emptyset = 0$, a statement of the form $|XF| = e \cdot$ `infDist` $X\,\ell$ could hold vacuously if ℓ were empty. Lemma 5 rules this out, since it computes the value explicitly.

Marden. Points are complex numbers. `focalSum f₁ f₂ z` $= |z - f_1| + |z - f_2|$, and the ellipse with foci f_1, f_2 and major axis $2A$ is $\{z : $ `focalSum` $f_1\, f_2\, z = 2A\}$.

3 Dandelin's theorem

3.1 The cone model

The classical proof uses one fact about spheres: two tangent segments from a point X to a sphere have the same length, since both have squared length $|XC|^2 - r^2$ by Pythagoras (`tangent_lengths_eq`). The second fact is specific to the cone.

Lemma 1. *If $X \in$ `cone V u k`, then $\|X - C_d\|^2 - (ds)^2 = (\|X - V\| - dk)^2$.*

So, for $d > 0$ and a non-apex point X of the upper nappe, the tangent length from X to the axis sphere with parameter d is $|\|X - V\| - dk|$, the distance along the generator from X to the circle at distance dk from the apex. This is the locus of contact: the intersection of the sphere with the hyperplane σ_d of Section 3.4. When E is finite-dimensional of dimension $N \geq 2$ and $d \neq 0$, it is a sphere of dimension $N - 2$ with centre $V + dk^2 u$ and radius $|d|sk$; for $N = 3$ it is a circle. The lower nappe is symmetric. We call `tangencyPoint V X k d` $= V + \frac{dk}{\|X-V\|}(X - V)$ the point where the generator through X touches the sphere. This point lies on the sphere, and the generator is perpendicular to the radius there (Lemma 19). As a consequence, if a plane through X touches the sphere at F, then $|XF| = |\|X - V\| - dk|$ (`dist_tangent_eq`).

3.2 The ellipse: existence, uniqueness and both directions

Theorem 2 (Dandelin, ellipse)**.** *Let $s < |m|$ and $mc > 0$. Then there are $0 < d_1 < d_2$ and points F_1, F_2 such that:*

(i) for every $d \in \mathbb{R}$, the sphere (C_d, ds) is tangent to the plane if and only if $d \in \{d_1, d_2\}$;

(ii) the two centres lie on opposite sides of the plane;

(iii) F_i lies in the plane and is the projection of C_{d_i} onto it;

(iv) for every X in the plane, $X \in$ `cone V u k` if and only if $|XF_1| + |XF_2| = (d_2 - d_1)k$.

```
theorem dandelin_exists {V u n p₀ : E} {k s : ℝ} (hu : ‖u‖ = 1) (hn : ‖n‖ = 1)
    (hks : k ^ 2 + s ^ 2 = 1) (hk : 0 < k) (hs : 0 < s) (hm : s < |⟪u, n⟫|)
    (hc : 0 < ⟪u, n⟫ * ⟪p₀ - V, n⟫) :
    ∃ d₁ d₂ : ℝ, ∃ F₁ F₂ : E, 0 < d₁ ∧ d₁ < d₂ ∧
      (∀ d : ℝ, |⟪(V + d • u) - p₀, n⟫| = d * s ↔ d = d₁ ∨ d = d₂) ∧
```

```
⟪(V + d₁ • u) - p₀, n⟫ * ⟪(V + d₂ • u) - p₀, n⟫ < 0 ∧
F₁ ∈ plane p₀ n ∧ F₂ ∈ plane p₀ n ∧
F₁ = (V + d₁ • u) - ⟪(V + d₁ • u) - p₀, n⟫ • n ∧
F₂ = (V + d₂ • u) - ⟪(V + d₂ • u) - p₀, n⟫ • n ∧
∀ X ∈ plane p₀ n, (X ∈ cone V u k ↔ ‖X - F₁‖ + ‖X - F₂‖ = (d₂ - d₁) * k)
```

The two hypotheses concern only the cone and the plane. The first, $s < |m|$, says that the plane is less steep than a generator. The second, $mc > 0$, says that the plane misses the apex and is on the side of the nappe `cone V u k`. Theorem 9 shows that together they characterise the ellipse regime.

Proof sketch. Flip n if necessary so that $m > 0$ and $c > 0$. The tangency condition $|c - dm| = ds$ is piecewise linear in d, and its solutions are $d_1 = c/(m + s)$ and $d_2 = c/(m - s)$ (`tangent_iff`). Take F_i to be the projections of the centres. For the forward direction of (iv), let X be a point of the section and write $t = \|X - V\|$. By Lemma 1, $|XF_1| + |XF_2| = |t - d_1k| + |t - d_2k|$. This equals $(d_2 - d_1)k$ if and only if $d_1k \le t \le d_2k$. Classically this "betweenness" is read off a picture. Here it is proved (`between_of_gram`). Apply the Cauchy–Schwarz inequality to the components of $X - V$ and of n orthogonal to u. Combined with the tangency equations $\langle X - V, n\rangle = d_1(m + s) = d_2(m - s)$, this gives $d_1k \le t \le d_2k$. For the converse, let $A = |XF_1|$, $B = |XF_2|$ and $p = \langle X - V, u\rangle$. For X in the plane, Pythagoras gives $A^2 = t^2 - 2d_1p + d_1^2k^2$, and similarly for B with d_2. Subtract the two equations and use $A + B = (d_2 - d_1)k$. This gives $kA = p - d_1k^2$, and hence $p > 0$ and $(p - kt)(p + kt) = 0$. So $p = kt$, which is the cone equation (`converse_alg`, `dandelin_converse`). □

Remark 3 (Methods). An earlier draft of the specification stated the bounded-section hypothesis as $|\langle n, u\rangle| > \cos\alpha$. Before the existence theorem was formalised, a numerical experiment on 2000 random cone–plane pairs (`code/check_phase2.py`) compared the prediction "every generator of the nappe meets the plane" with the candidate inequality. It showed that the correct condition is $|\langle n, u\rangle| > \sin\alpha$. A generator direction $w = ku + se$ with $e \perp u$ a unit vector has $\langle w, n\rangle$ ranging over $[km - s\sqrt{1 - m^2},\ km + s\sqrt{1 - m^2}]$, and this interval avoids 0 exactly when $m^2 > s^2$. With the $\cos\alpha$ hypothesis, the existence statement is false for $\alpha > \pi/4$, where it admits parabolic and hyperbolic sections. A failed proof attempt would not have revealed this as directly.

3.3 The hyperbola

In the hyperbola regime the plane cuts both nappes, so the statement has to be about the double cone. Take one sphere in each nappe, that is, axis spheres with parameters of opposite signs.

Theorem 4 (Dandelin, hyperbola)**.** *Let $|m| < s$ and $c \neq 0$. Then $e := \sqrt{1 - m^2}/k > 1$, and there are $d_2 < 0 < d_1$ and points F_1, F_2 such that:*

(i) for every $d \in \mathbb{R}$, the sphere $(C_d, |d|s)$ is tangent to the plane if and only if $d \in \{d_1, d_2\}$, and F_i is the projection of C_{d_i} onto the plane;

(ii) for every X in the cutting plane, $|XF_2| - |XF_1| = D$ if X is on the upper nappe, and $|XF_1| - |XF_2| = D$ if X is on the lower nappe, where $D = (d_1 - d_2)k$;

(iii) for every X in the plane, $X \in$ `cone2 V u k` if and only if $\big||XF_1| - |XF_2|\big| = D$;

(iv) every X of the section satisfies $|XF_i| = e \cdot \operatorname{dist}(X, \ell_i)$ for $i = 1, 2$, where ℓ_i is the directrix of Section 3.4.

This is `dandelin_hyperbola`. Its orientation step flips n so that $c > 0$, and then $d_1 = c/(m+s)$ and $d_2 = c/(m-s)$ as before. The closed form is $d_1 - d_2 = 2sc/(s^2 - m^2)$ (`hyp_param_sub`). The proof of (ii) needs the analogue of betweenness: a point of the upper nappe in the plane lies beyond the upper circle of contact, $d_1 k \le \|X - V\|$ (`hyp_between`). The proof again uses Cauchy–Schwarz. The lower-nappe identity follows from the upper one under the symmetry $u \mapsto -u$, $d_i \mapsto -d_{3-i}$.

For the converse (`hyp_converse`), write $||XF_1| - |XF_2|| = D$ as $|XF_1| - |XF_2| = \varepsilon D$ with $\varepsilon = \pm 1$. The same subtraction as in the ellipse case gives $\varepsilon k A = d_1 k^2 - p$, and then $p^2 = k^2 t^2$, which is the equation of `cone2`. The sign ε tells us which nappe X lies on, so no positivity argument is needed. The converse cannot be stated for one nappe. Each branch of the hyperbola lies on its own nappe, and the absolute value is necessary.

3.4 Focus and directrix; the parabola

Let $\sigma_d = \{Y : \langle Y - V, u\rangle = dk^2\}$ be the plane perpendicular to the axis at height dk^2. The circle of contact of the sphere with parameter d lies in σ_d (`tangencyPoint_mem_axisPlane`). The *directrix* is $\ell_d = \texttt{directrix V u n p}_0\,(dk^2) = \texttt{plane p}_0\ \texttt{n} \cap \sigma_d$. The eccentricity is

$$e = \texttt{ecc}\ m\ k = \frac{\sqrt{1-m^2}}{k} = \frac{\cos\beta}{\cos\alpha}.$$

Lemma 5. *If $m^2 < 1$ and X lies in the plane, then `infDist` $X\,\ell_d = |\langle X - V, u\rangle - dk^2|/\sqrt{1-m^2}$.*

The proof exhibits the nearest point $X - \frac{\delta}{1-m^2}(u - mn)$ explicitly, where $\delta = \langle X - V, u\rangle - dk^2$, and uses Cauchy–Schwarz against $u - mn$ for the lower bound. It holds in every inner product space. In particular ℓ_d is non-empty whenever the plane is.

Theorem 6 (focus–directrix)**.** *Suppose $m^2 < 1$, X lies in the plane and on `cone V u k`, and a sphere (C_d, ds) touches a plane through X at F. Then $|XF| = e \cdot$ `infDist` $X\,\ell_d$. Moreover $e < 1$ iff $s < |m|$, $e = 1$ if $s = |m|$, and $e > 1$ iff $|m| < s$. In the ellipse regime (the hypotheses of Theorem 2 together with $m^2 < 1$) both foci satisfy this relation with their own directrices.*

The proof composes two elementary facts. The first is $|XF| = \operatorname{dist}(X, \sigma_d)/\cos\alpha$ (`focal_dist_eq_infDist_axisPlane`): the tangent length along the generator is $1/\cos\alpha$ times its axial component. The second is $\operatorname{dist}(X, \sigma_d) = \cos\beta \cdot \operatorname{dist}(X, \ell_d)$ for X in the plane (`infDist_axisPlane_eq_mul`). The hypothesis $m^2 < 1$ excludes the plane perpendicular to the axis. There the section is a circle and ℓ_d is empty.

Theorem 7 (parabola)**.** *Let $s = |m|$ and $mc > 0$. Then $e = 1$, and there is exactly one $d > 0$ for which (C_d, ds) is tangent to the plane. Its tangency point F satisfies $|XF| =$ `infDist` $X\,\ell_d$ for every X in the section (`directrix_parabola`).*

The converse of the focus–directrix property also holds, and it turns out to be a statement about the double cone. Let F be the tangency point of $(C_d, |d|s)$ with the plane and let X be any point of the plane. Combining Lemma 5 with Pythagoras (`focal_sq`) gives

$$(k|XF|)^2 - \big(\langle X - V, u\rangle - dk^2\big)^2 = k^2\|X - V\|^2 - \langle X - V, u\rangle^2.$$

So the relation $|XF| = e\operatorname{dist}(X, \ell_d)$ restricted to the plane *is* the equation of `cone2`.

Theorem 8 (converse)**.** *If $m^2 < 1$, X is in the plane, the plane touches $(C_d, |d|s)$ at F and $|XF| = e \cdot$ `infDist` $X\,\ell_d$, then $X \in$ `cone2 V u k` (`directrix_converse`; the formal lemma needs only the pointwise conditions $|FC_d| = |d|s$ and $\langle X - F, F - C_d\rangle = 0$, not tangency of the whole plane, and does not assume $s > 0$). Moreover:*

(a) *in the ellipse and parabola regimes, the section equals the one-nappe locus:* $X \in$ `cone V u k` *if and only if the relation holds, for each focus (*`directrix_iff_ellipse`*,* `directrix_iff_parabola`*);*

(b) *in the hyperbola regime,* $X \in$ `cone2 V u k` *if and only if the relation holds, for each focus. The one-nappe version is false: there is a point* Y *of the plane that satisfies the relation for* both *foci but is not on* `cone V u k`*, and there is another that is not on* `cone V (-u) k` *(*`directrix_iff_hyperbola`*).*

The counterexamples in (b) are proved in Lean, not merely observed. The witnesses are the section points on the other nappe constructed in Theorem 9(c). They are not the apex, because $c \neq 0$ and a point lying on both nappes must be the apex (`eq_of_mem_both_nappes`).

3.5 Regimes and non-emptiness

So far the conditions $s < |m|$, $s = |m|$ and $|m| < s$ have only been hypotheses. The following theorem shows that they have the intended geometric meaning. All of its parts assume the standing hypotheses on u, n, k, s and $c \neq 0$.

Theorem 9 (regimes)**.**

(a) *If* $s \leq |m|$*, the section of the double cone lies in a single nappe: in* `cone V u k` *if* $mc > 0$ *and in* `cone V (-u) k` *if* $mc < 0$ *(*`one_nappe`*; any* E*).*

(b) *If* E *is finite-dimensional and* $\dim E \geq 3$*, then* $s < |m|$ *holds iff* `plane` $\cap$ `cone2` *is bounded (*`bounded_iff`*), iff it lies in one nappe and is bounded (*`ellipse_regime_iff`*). The forward direction holds in any* E *(*`ellipse_regime`*, via* `section_bounded`*).*

(c) $|m| < s$ *holds iff the plane meets both nappes,* `cone V u k` *and* `cone V (-u) k` *(*`hyperbola_regime_iff`*; any* E*).*

(d) *If* E *is finite-dimensional,* $\dim E \geq 2$ *and* $s < |m|$*,* $mc > 0$*, then the section is non-empty (*`ellipse_nonempty`*). The witness is the intersection of the plane with a generator of the nappe.*

(e) *If* E *is finite-dimensional,* $\dim E \geq 3$ *and* $|m| = s$*, the section lies in one nappe and is unbounded (*`parabola_regime`*), and in particular it is non-empty.*

The proofs give explicit geometric witnesses. (This is not a claim about computational content: like most of Mathlib, the developments use `Classical.choice`.) A generator $V + t(ku + se)$ with $e \perp u$ a unit vector meets the plane at $t = c/\langle ku + se, n\rangle$, and the sign of t decides the nappe (`gen_point`). For the hyperbola, the directions $e = \pm q/\|q\|$ with $q = n - mu$ give parameters of opposite signs. For unboundedness, choose a unit $f \perp u, n$ (this is where finite dimension and $\dim E \geq 3$ are used, `exists_unit_orth`) and tilt e in the (q, f)-plane so that $\langle ku + se, n\rangle$ becomes an arbitrarily small nonzero number (`section_unbounded`).

The dimension hypotheses are necessary. If $\dim E = 1$, the cone is $\{V\}$, so the section is empty. If $\dim E = 2$, a "plane" is a line and the double cone is a pair of lines through V, so every section is bounded, even when $|m| \leq s$. These two remarks are elementary and are not formalised. `regime_R3` bundles (b), (c) and (d) for `EuclideanSpace` $\mathbb{R}$ `(Fin 3)`.

3.6 Planes through the apex

Theorem 10. *Let $c = 0$ and put $a = u - mn$ (`apexDir u n`).*

(a) If $s < |m|$, then `plane` $\cap$ `cone2` $= \{V\}$ (any E).

(b) If $|m| = s$, then $a \neq 0$ and the section is the line $\{V + ta\}$ (any E).

(c) If $|m| < s$ and $\dim E = 3$, then there are a unit vector $f \perp u, n$ and $\rho > 0$ such that the section is the union of the two lines $\{V + t(a \pm \rho f)\}$, and these lines meet only at V.

Here $\rho = \sqrt{(1 - m^2)(s^2 - m^2)}/k$ (`apexRho`). In case (b), equality in Cauchy–Schwarz forces $X - V$ to be parallel to a. In case (c), $\dim E = 3$ is used to decompose every $w \perp n$ in the orthogonal pair a, f (`decomp_of_finrank_le_three`). Four nonzero pairwise orthogonal vectors would be linearly independent. Then the cone equation factors as $k^2(\beta - \alpha\rho)(\beta + \alpha\rho) = 0$, where α and β are the coordinates of w along a and f. If E is finite-dimensional with $N = \dim E > 3$, the section is not two lines but a quadric cone in the $(N - 1)$-dimensional cutting hyperplane, of locus dimension $N - 2$. (For example, in $\mathbb{R}^4$ with $u = e_4$, $n = e_1$ and $k = s = 1/\sqrt{2}$, the section is $\{x_1 = 0,\ x_4^2 = x_2^2 + x_3^2\}$, a two-dimensional cone in a three-dimensional hyperplane.) This description is informal. Only the containment of the two lines in the section is formalised (`apex_two_lines_subset`, for E finite-dimensional with $\dim E \geq 3$).

4 Marden's theorem

4.1 Statement

Let `cubic a b c` $= (X - a)(X - b)(X - c) \in \mathbb{C}[X]$. We state the critical points through Mathlib's formal derivative. The hypothesis is

$$\texttt{derivative (cubic a b c)} = 3(X - z_1)(X - z_2),$$

that is, z_1, z_2 are the roots of p' counted with multiplicity. `vieta` derives $z_1 + z_2 = \frac{2}{3}(a + b + c)$ and $z_1 z_2 = \frac{1}{3}(ab + bc + ca)$ from it. Non-collinearity is stated as $\operatorname{Im} \frac{c-a}{b-a} \neq 0$.

Theorem 11 (Marden)**.** *Let a, b, c be distinct and non-collinear, and let z_1, z_2 be as above. Put $2A =$ `focalSum` $z_1\ z_2\ \frac{a+b}{2}$. Then:*

(i) all three midpoints lie on the ellipse $\{z : |z - z_1| + |z - z_2| = 2A\}$;

(ii) for each side xy and every real $t \neq 0$, $2A <$ `focalSum` $z_1\ z_2\ \left(\frac{x+y}{2} + t(y - x)\right)$;

(iii) $\frac{z_1+z_2}{2} = \frac{a+b+c}{3}$.

```
theorem marden {a b c z₁ z₂ : ℂ}
    (hder : derivative (cubic a b c) = C 3 * (X - C z₁) * (X - C z₂))
    (hab : a ≠ b) (hbc : b ≠ c) (hca : c ≠ a)
    (hnc : ((c - a) / (b - a)).im ≠ 0) :
    let A := focalSum z₁ z₂ ((a + b) / 2) / 2
    (focalSum z₁ z₂ ((a + b) / 2) = 2 * A ∧
      focalSum z₁ z₂ ((b + c) / 2) = 2 * A ∧
      focalSum z₁ z₂ ((c + a) / 2) = 2 * A) ∧
    (∀ t : ℝ, t ≠ 0 → 2 * A < focalSum z₁ z₂ ((a + b) / 2 + t * (b - a))) ∧
    (∀ t : ℝ, t ≠ 0 → 2 * A < focalSum z₁ z₂ ((b + c) / 2 + t * (c - b))) ∧
    (∀ t : ℝ, t ≠ 0 → 2 * A < focalSum z₁ z₂ ((c + a) / 2 + t * (a - c))) ∧
    (z₁ + z₂) / 2 = (a + b + c) / 3
```

The distinctness hypotheses follow from `hnc`. We keep them for readability.

4.2 Why metric tangency is tangency

Condition (ii) says that the side line through the midpoint m meets the closed region $K = \{z : \texttt{focalSum}\ z_1\ z_2\ z \le 2A\}$ only at m. The ellipse is non-degenerate: $2A > |z_1 - z_2|$ by Lemma 22. So K is a compact, strictly convex set with non-empty interior, and its boundary is a smooth curve. For such a set, a line through a boundary point m that is not tangent at m enters the interior of K on one side of m, so it contains points with focal sum $< 2A$. The tangent line at m supports K and, by strict convexity, meets K only at m. Hence (ii) is equivalent to "the side line is tangent to the ellipse at the midpoint". For *inscribed*, argue as follows. By (ii), K lies in one of the two closed half-planes bounded by each side line. The midpoint of another side lies in K and strictly inside the triangle-facing half-plane, so that is the half-plane containing K. Intersecting the three half-planes, K lies in the closed triangle. The tangency equivalence is informal, standard convex geometry. We have not formalised it, because Mathlib has no notion of an ellipse or of its tangent lines. The containment argument was informal in version 1. The Layer-15 file now proves that the Marden ellipse lies in the closed triangle (`Marden.Classical.l15_marden_steiner`). It also proves that if a side line meets the curve $\{\texttt{focalSum} = 2A\}$ only at the midpoint, then (ii) holds for that side (`Marden.Classical.l15_strict_of_inter`); see Section 7.1. The formal *statement* avoids derivatives and angles entirely; the uniqueness *proof* (Section 4.4) does use a derivative. Section 4.4 also shows that the weaker "supporting line" condition ($\le$ instead of $<$) already determines the ellipse.

4.3 The key identity

Lemma 12. *Let z_1, z_2 be the critical points as in Section 4.1 (so the Vieta relations hold), and let $m = \frac{a+b}{2}$. Then $(m - z_1)(m - z_2) = \frac{1}{3}p'(m) = -\frac{(b-a)^2}{12}$.*

Proof. In $p'(m) = (m-b)(m-c) + (m-a)(m-c) + (m-a)(m-b)$, the first two terms cancel because $m - a = -(m - b)$. The third term is $-(b-a)^2/4$. Formally, the lemma is a single `linear_combination` of the two Vieta relations. □

So $(m - z_1)(m - z_2)/(b - a)^2 = -1/12$ is a *negative real*. Geometrically, the side direction $d = b - a$ makes equal angles with $m - z_1$ and $z_2 - m$, which is the reflection law of the ellipse at m. The formal argument turns this into an inequality.

Lemma 13. *If $d \neq 0$ and $uv = -d^2/12$, then $\|u\| + \|v\| \le \|u + td\| + \|v + td\|$ for every real t. The inequality is strict for $t \neq 0$ when $\mathrm{Im}(u/d) \neq 0$.*

Proof. Write $u = wd$. The hypothesis forces $v = -\mu\bar{w}d$ with $\mu = 1/(12|w|^2) > 0$. Hence $\|u + td\| + \|v + td\| = |d|\,(|w + t| + |\mu w - t|) \ge |d|(1 + \mu)|w|$ by the triangle inequality, with equality at $t = 0$. When $\mathrm{Im}\, w \neq 0$ and $t \neq 0$, the two summands are not on the same ray, and Mathlib's strict triangle inequality (`norm_add_lt_of_not_sameRay`) applies. □

Non-collinearity implies that the focus is off the side line, $\mathrm{Im}\,\frac{m-z_1}{b-a} \neq 0$ (`focus_off_side`). This gives (ii) for the side ab. The other sides follow by cyclic relabelling.

Lemma 14. *With z_1, z_2 the critical points as in Section 4.1, $\left(\texttt{focalSum}\ z_1\ z_2\ \frac{a+b}{2}\right)^2 = \frac{1}{2}|z_1 - z_2|^2 + \frac{1}{9}\left(|a-b|^2 + |b-c|^2 + |c-a|^2\right)$.*

The proof combines the parallelogram law with $|m - z_1|\,|m - z_2| = |b - a|^2/12$. The right-hand side is symmetric in a, b, c, so all three midpoints have the same focal sum. This is (i). It needs only the Vieta relations, not non-collinearity. Part (iii) is Vieta.

4.4 Uniqueness

Theorem 15 (uniqueness)**.** *Let a, b, c, z_1, z_2 be as in Theorem 11, and let $f_1, f_2 \in \mathbb{C}$ and $A' \in \mathbb{R}$ be arbitrary. Suppose that the three midpoints satisfy* `focalSum` $f_1\, f_2 = 2A'$ *and that each side line meets* $\{$`focalSum` $f_1\, f_2 \le 2A'\}$ *only at its midpoint. Then $\{f_1, f_2\} = \{z_1, z_2\}$ and $A' = A$ (*`steiner_unique`*). The same conclusion holds if the strict inequality is weakened to $2A' \le$* `focalSum` *along each side (the side is a supporting line) and the ellipse is assumed non-degenerate, $|f_1 - f_2| < 2A'$ (*`steiner_inellipse_unique`*).*

The non-degeneracy premise of the weak form is redundant: it already follows from the three midpoint equalities and non-collinearity (`nondeg_of_midpoints`, Lemma 22), and `steiner_unique` discharges it in exactly this way. It remains an explicit premise of `steiner_inellipse_unique` only as a matter of how that statement is phrased. So weakening $<$ to $\le$ costs no additional geometric restriction.

Corollary 16. *For a non-degenerate triangle, a focal ellipse passes through the three midpoints with each side touching it only there if and only if its foci are the critical points of p and its major axis is $2A$ (*`steiner_inellipse_iff`*).*

The route has three steps.

Lemma 17 (converse tangency)**.** *If $d \neq 0$, $\|u - v\| < \|u\| + \|v\|$, and $t = 0$ minimises $t \mapsto \|u + td\| + \|v + td\|$, then $uv = -\lambda d^2$ with $\lambda = \|u\|\|v\|/\|d\|^2 \ge 0$.*

The proof sets the derivative at the minimum to zero (`IsLocalMin.hasDerivAt_eq_zero`). A factorisation then leaves two cases. One is the reflection law. The other puts u and $-v$ on the same ray, which contradicts non-degeneracy.

Lemma 18. *Let a, b, c be distinct and non-collinear, and let m_i, d_i be their cyclic side midpoints and directions ($m_1 = \frac{a+b}{2}$, $d_1 = b - a$, and cyclically). Suppose the reflection law $(m_i - f_1)(m_i - f_2) = -\lambda_i d_i^2$ holds at all three midpoints, with $\lambda_i \ge 0$, and the three focal sums are equal. Then $f_1 + f_2 = \frac{2}{3}(a + b + c)$, $f_1 f_2 = \frac{1}{3}(ab + bc + ca)$ and $\lambda_1 = \lambda_2 = \lambda_3 = \frac{1}{12}$ (*`foci_of_reflection`*).*

Equal focal sums are genuinely needed: the reflection laws with $\lambda_i \ge 0$ alone do not determine the foci. For example, let $a = 0$, $b = 4$, $c = 1+3i$ and let f_1, f_2 be the roots of $z^2 - (3+4t+2i)z + (\frac{1}{2} + 10t+4i)$ for real t. Then all three reflection equations hold with $\lambda_1 = \frac{3}{32} - \frac{t}{8}$, $\lambda_2 = \frac{1}{9} - \frac{t}{3}$, $\lambda_3 = t$, which are nonnegative for $0 \le t \le \frac{1}{3}$. Only $t = \frac{1}{12}$ gives equal midpoint focal sums and the Marden foci (by the lemma, since then $\lambda_i = \frac{1}{12}$). The reflection identities and the nonnegativity interval are checked exactly in `code/check_family.py`; uniqueness of the equal-sum member follows from the preceding lemma. Squaring through $\|u\|\|v\| = \lambda\|d\|^2$ turns each equality of focal sums into a real linear condition. An explicit `linear_combination` certificate shows that it forces $\lambda_i = \lambda_j$ (`lam_eq_core`). The certificate multiplies $(\lambda_1 - \lambda_2)$ by the square of a quantity that is nonzero exactly under non-collinearity. Once all the λ_i equal λ, the combination $\sum_{\mathrm{cyc}}(m_2 - m_3)\big[(m_1 - f_1)(m_1 - f_2) + \lambda d_1^2\big]$ is independent of f_1, f_2 and equals $(a-b)(a-c)(b-c)(12\lambda - 1)/8$ (`lam_twelfth_core`). So $\lambda = \frac{1}{12}$ is a consequence, not an assumption. The sum and product of the foci then follow linearly, and $(f_1 - z_1)(f_1 - z_2) = 0$ identifies them.

Scope of uniqueness. The competitors are all sets $\{z : |z - f_1| + |z - f_2| = 2A'\}$ with arbitrary f_1, f_2, A'. Every non-degenerate ellipse in the plane, including every circle, has this form, and the degenerate cases are excluded automatically (Lemma 22). The classical statement that the Steiner inellipse is the unique non-degenerate *conic* tangent to the sides at their midpoints is not formalised. Stating it would require a defined notion of non-degenerate conic and of tangency to it, and a bridge showing that a conic of elliptic type has a focal representation. Mathlib has neither.

5 Does the statement mean what it says?

A formal theorem can be true for the wrong reason: its hypotheses may be unsatisfiable, or its definitions may not mean what the reader thinks. The lemmas below address the points where we see a risk.

Lemma 19 (inscribed spheres)**.** *Let $X \in$ `cone V u k` with $X \neq V$, and let $ds \geq 0$. Then $T =$ `tangencyPoint V X k d` lies on the sphere (C_d, ds), and $X - T \perp T - C_d$. Moreover no point of the cone lies in the open ball: $(ds)^2 \leq \|X - C_d\|^2$.*

So an axis sphere with $d > 0$ really is inscribed in the cone. It touches every generator and never crosses the surface. The condition "tangent to the plane" in Theorems 2 and 4 is the distance condition $|\langle C - p_0, n\rangle| = r$. The next lemma connects it to the pointwise notion.

Lemma 20. *If $F = C - \langle C - p_0, n\rangle\, n$ and $|\langle C - p_0, n\rangle| = r$, then F lies in the plane, $\|F - C\| = r$, and $X - F \perp F - C$ for every X in the plane (`tangent_of_proj`, with `proj_mem_plane`).*

Remark 21. Mathlib has a notion of tangency, `EuclideanGeometry.Sphere.IsTangentAt`, for a sphere and an affine subspace. `dist_eq_of_isTangentAt` proves the equal-tangents property in that language. `dandelin_of_between_isTangentAt` is a version of the ellipse theorem that uses Mathlib spheres and `IsTangentAt`, with betweenness as a hypothesis. The existence theorems of Section 3 use the projection form instead. In the main files we have not restated them with `IsTangentAt`. The Layer-14 file restates them with Mathlib spheres, `IsTangent` and `IsTangentAt` (Section 7.1).

Bridge lemmas. The module `Dandelin/Bridge.lean` ties the definitions to the classical notions. For $k = \cos\alpha$ with $0 \leq \alpha \leq \pi$ and $X \neq V$, `mem_cone_iff_angle` says that $X \in$ `cone V u k` iff the generator VX makes angle α with u, and `mem_cone2_iff_angle` gives angle α or $\pi - \alpha$ for `cone2`. `generator_mem_cone` and `exists_generator_of_mem_cone` give the parametric form $V + t\,w$ with $t \geq 0$, $\|w\| = 1$, $\langle w, u\rangle = k$. `mem_plane_iff` is the normal equation, and `plane_eq_mk'` identifies `plane` with Mathlib's affine subspace through p_0 with direction $n^\perp$. `tangencyPoint_mem_line` and `isTangentAt_tangencyPoint` show that the generator line VX is tangent, in the sense of `IsTangentAt`, to the axis sphere at `tangencyPoint`. `isTangentAt_focus`$_1$ and `isTangentAt_focus`$_2$ show that the two Dandelin spheres touch the cutting plane, in the same sense, exactly at `focus`$_1$ and `focus`$_2$.

Non-vacuity. The hypotheses of Theorems 2 and 4 are satisfiable in the intended three-dimensional setting. Non-emptiness is established with the following scopes. Theorem 9(c) proves that the hyperbola section has points on both nappes (in any E), (d) that the ellipse section is non-empty when E is finite-dimensional with $\dim E \geq 2$, and (e) the same for the parabola when E is finite-dimensional with $\dim E \geq 3$. For non-circular sections ($m^2 < 1$) the directrices are non-empty by Lemma 5. In version 2 the Layer-14 witness theorems also prove the hypotheses on explicit configurations in $\mathbb{R}^3$ and $\mathbb{C}$ (Section 7.1). For Marden:

Lemma 22. *For all $a, b, c \in \mathbb{C}$ there are z_1, z_2 with $p' = 3(X - z_1)(X - z_2)$; this is* `exists_critical_points`*. If a, b, c are non-collinear and the three midpoints have common focal sum $2A'$ with respect to some f_1, f_2, then $|f_1 - f_2| < 2A'$ (*`nondeg_of_midpoints`*).*

The second part applies in particular to $f_i = z_i$, so the Marden ellipse is not degenerate. If it were a segment or a point, the three midpoints would lie on one line, and then so would a, b, c.

Sharpness. The counterexample of Theorem 8(b) is a proved statement. It shows that the hyperbola converse cannot be stated for a single nappe.

The author has checked each statement in Table 1 against its informal counterpart by hand.

6 Related work

Mathlib. At tag `v4.33.1`, `docs/1000.yaml` lists "Dandelin's theorem" and "Marden's theorem" with no declaration or author, and the same holds on the master branch as of 30 September 2026 [9]. A case-insensitive search of the Mathlib sources at that tag for `dandelin`, `marden`, `inellipse`, `ellipse` and "conic section" found nothing. Mathlib does contain the Gauss–Lucas theorem (`Mathlib/Analysis/Complex/Polynomial/GaussLucas.lean`), which also locates critical points, and the sphere tangency notion mentioned in Remark 21.

Isabelle. The Archive of Formal Proofs has an entry by Eberl on the geometry of the critical points of complex polynomials [3]. It covers the Gauss–Lucas and Jensen theorems. Its bibliography cites Marden's book, but its theory file does not mention Marden's theorem or ellipses. We searched the titles and abstracts of all 1029 entries in the AFP metadata (mirror repository, commit `7de2cdd`, 29 September 2026) for Marden, Dandelin, inellipse, ellipse and conic section, and found no match. This search did not cover the full text of every theory.

HOL Light. A search of the HOL Light repository [5] (commit `cba9198`, 16 September 2026) for `marden`, `dandelin`, `inellipse` and `ellipse` returned no matches. Its `100/pascal.ml` treats conics, but only through Pascal's theorem [1].

Other sources. The development reports record GitHub searches: code search for `Dandelin`, `Marden`, `inellipse` and "Steiner ellipse" restricted to Lean, repository search, and a search of Mathlib issues. The only relevant hit was Mathlib issue #6091, a wish-list that mentions Marden's theorem. Web searches for formalisations in Lean, Isabelle, Rocq and HOL found only informal expositions. One search-engine snippet attached to a figure page on ResearchGate juxtaposes "the configuration of Marden's theorem" with the sentence "We also provide a machine-checked Lean 4 formalization of the main result".[1] The page was not accessible to us, and we could not determine what that formalisation concerns. A search of the Lean Zulip chat for "Dandelin" and "Marden" on 30 September 2026 returned no relevant results. We did not search the Mizar library, and Rocq was covered only by web search. We therefore claim only that we found no earlier complete formalisation of either theorem in the sources searched.

[1] Figure page `https://www.researchgate.net/figure/A-typical-trajectory-of-the-vector-field-in-4-and-the-level-sets-of-the-Lyapunov_fig1_323846042`. Its title concerns a vector field and a Lyapunov function; fetching it returned HTTP 403. We record it as an unresolved lead, not as evidence of prior work or of its absence.

7 Verification

Both libraries were built on a separate machine with Lean `v4.33.1`, using prebuilt Mathlib artefacts. Excluding Mathlib, the full Dandelin build takes about 10 s and the Marden build about 17 s, with no warnings. A grep over the library sources (not the scratch files) for `sorry`, `admit`, `native_decide`, `axiom`, `#eval`, `run_cmd`, `set_option`, `macro`, `elab`, `syntax`, `notation` and `import Lean` finds nothing. We repeated this grep on the source mirror used for this paper.

Axioms were checked with `#print axioms` in scratch files outside the libraries, for 63 distinct Dandelin declarations (seven files `scratch/Ax*.lean`) and 11 Marden declarations. Each reports only `propext`, `Classical.choice` and `Quot.sound`. The queries cover every row of Table 1. Five rows (`cone_tangent_sq`, `tangent_of_proj`, `focalSum_midpoint_sq`, `focal_lt_of_mul`, `vieta`) are covered transitively, because `#print axioms` follows dependencies and queried theorems use them. `cone_outside_sphere` is used by no other library theorem, so it has its own query (`scratch/AxConeOutside.lean`). The evidence is of two kinds. *Stored logs*: the outputs for 27 of the Dandelin queries (`scratch/ax5.out`, `scratch/ax6.out`, `scratch/ax_cone_outside.out`) and for all 11 Marden queries (`ax.log`) are kept as files in the repositories. *Report-only*: the outputs for the remaining 36 Dandelin queries (the first four phases, `scratch/Ax.lean`–`Ax4.lean`) were recorded only in the development reports; the query files are kept and can be rerun. Both repositories also have a `gate.sh` that rejects forbidden tokens, builds the library, generates its own `#print axioms` queries for every declaration cited in this paper and every query file in `scratch/` (85 declarations for Dandelin, 18 for Marden), and fails on any other axiom. On the published commits both print `PASS`. In the version-2 commits the gates also list the Layer-14 and Layer-15 declarations (158 for Dandelin, 56 for Marden). Both print `PASS`, and every listed declaration reports only the three standard axioms. This regenerates, on the final sources, the 36 Dandelin reports that were previously recorded only in the development notes. `tests/tamper.sh` plants six kinds of defect in scratch copies (a `sorry`, a split `axiom`, a fake `#eval` report, a macro, a weakened statement, a wrapped extra axiom); in both repositories the gate rejects all six, each with a nonzero exit status and the expected failure message. The same holds for the version-2 commits.

Before formalising each statement, we tested it numerically. These checks produced the correction in Remark 3. The scripts are in `code/`:

- `check_dandelin.py`, `check_phase2.py`, `check_phase3.py`: focal sums, the regime prediction (2000/2000 agreement), the converse, and the focus–directrix relation on 34 231 points over 3000 cone–plane pairs (maximum relative error $6.8 \cdot 10^{-12}$);
- `check_phase4.py`: the hyperbola constant and its converse on 489 890 points;
- `regime_check.py`, `degenerate_check.py`: the constructions of Theorem 9, the key identity for the directrix converse, the one-nappe counterexample (69 106 cases) and the apex sections, all with zero failures;
- `check_marden.py`, `check_unique.py`: Lemma 12 symbolically, Lemma 14 and the tangency on 2000 random triangles, both `linear_combination` certificates of Section 4.4 symbolically, and a sampled perturbation test (a necessary-condition check for uniqueness, not a verification of it);
- `check_family.py`: exact SymPy check that the one-parameter family of Section 4.4 satisfies the reflection equations with nonnegative λ_i (the focal sums are only sampled).

For this paper we re-ran all of the scripts except `check_phase4.py`, which takes about nine minutes. The outputs agreed with the recorded ones. The numerical checks are evidence that the statements are the right ones. They play no role in the proofs.

The step that matters most, comparing the formal statements with the informal ones, cannot be automated. Sections 2 and 5 are meant to make that comparison easier.

7.1 Layers 14 and 15

Version 2 adds a Layer-14 and a Layer-15 library file to each repository: `Dandelin/L14.lean`, `Marden/L14.lean`, `Dandelin/L15Classical.lean` and `Marden/L15Classical.lean`. The Layer-15 namespaces are `Dandelin.Classical` and `Marden.Classical`. The gates check selected Layer-14 declarations and the Layer-15 proof declarations, not every declaration in these files. Every gate-listed declaration reports only the three standard axioms. The pre-existing proof modules are unchanged; each root module gains two imports.

Layer 14: statements in Mathlib's language, with witnesses. Nine main results are restated with Mathlib-only types and proved from the library theorems through new bridges:

- seven Dandelin results:
  - Theorem 2: `Dandelin.l14_dandelin_ellipse`, in any real inner product space;
  - Theorem 4 together with the double-cone equivalence of Theorem 8(b): `Dandelin.l14_dandelin_hyperbola`;
  - Theorems 6 and 8(a) for the ellipse: `Dandelin.l14_directrix_ellipse`;
  - Theorems 7 and 8(a) for the parabola: `Dandelin.l14_parabola`;
  - Theorem 9(b)–(d) in $\mathbb{R}^3$: `Dandelin.l14_regime_R3`;
  - Theorem 10 in $\mathbb{R}^3$: `Dandelin.l14_apex_R3`;
  - Lemma 19: `Dandelin.l14_inscribed`;
- two Marden results: Theorem 11 (`Marden.l14_marden`) and Theorem 15 in its strict form (`Marden.l14_steiner_unique`).

The Mathlib-only types are as follows.

- The cone is defined by angles with `InnerProductGeometry.angle`, with the apex included explicitly, because Mathlib sets the angle of 0 to $\pi/2$.
- The plane is `AffineSubspace.mk'` $p_0\,(\mathbb{R}n)^{\perp}$.
- Spheres and tangency are `EuclideanGeometry.Sphere`, `IsTangent` and `IsTangentAt`.
- The directrix is the intersection of two affine subspaces, and distances to it use `Metric.infDist`.
- For Marden we use `¬ Collinear`, `Polynomial.derivative`, `midpoint`, `line[ℝ,x,y]` and `Finset.centroid`.

A mechanical check of the elaborated types, with notation turned off, found no constant from `Dandelin` or `Marden` in any restated statement.

Witness theorems prove every hypothesis on an explicit configuration, together with a non-degenerate conclusion:

- `Dandelin.l14w_dandelin_ellipse`: $\mathbb{R}^3$ with $\cos\alpha = 4/5$; $F_1 \neq F_2$, focal constant > 0, a section point other than the apex, and non-empty directrices;
- `Dandelin.l14w_dandelin_hyperbola`: $F_1 \neq F_2$ and section points on both nappes;
- `Dandelin.l14w_parabola`: a section point other than the apex;
- `Dandelin.l14w_regime_inscribed`: a section point other than the apex, at angle α;
- `Dandelin.l14w_apex`: the point, line and two-line cases, with distinct non-apex points on each line;
- `Marden.l14w_marden`: the triangle $0, 5, 1+4i$ with foci $1+2i$ and $3+\frac{2}{3}i$ (the derivative identity is `Marden.l14w_derivative`). It proves distinct vertices, distinct foci, distinct midpoints and a non-degenerate ellipse.

Not restated:

- Corollary 16, which is the conjunction of the two Marden restatements;
- the weak form of Theorem 15;
- Theorem 9(a) and (e), and the versions in general dimension;
- the one-nappe counterexample of Theorem 8(b);
- the internal lemmas. Lemma 20 has a Mathlib-only form as the bridge `Dandelin.l14_isTangent_plane_iff`.

No restatement turned out false or needed an extra hypothesis.

Layer 15: comparison with textbook statements. Another AI agent collected the statements of the classical theorems from the literature, with citations: Dandelin, Hilbert and Cohn-Vossen, Kalman, Marden, Siebeck, and standard reference pages. An AI model was given only these textbook statements. It saw no code and no part of this paper, and its transcript was audited. It wrote 13 Lean statements with Mathlib notions and its own definitions: 12 theorems and one well-formedness obligation, namely that the directrix is a line. Its Dandelin sphere is *any* sphere of positive radius meeting every generator in exactly one point and the plane in exactly one point. It marked three ambiguities and no extra hypotheses.

A comparer copied these statements verbatim. For each one it either proved it from the development or classified it. It added no hypotheses and left the library unchanged. The outcome: 11 covered, 1 narrower, 1 out of scope, and none false as written.

- *Covered and proved* (11): `cov_Conics1_a`, `cov_Conics1_b`, `cov_Conics2_a`, `cov_Conics2_b`, `cov_Conics3_obligation` (the obligation is proved, not assumed), `cov_Conics3_a`, `cov_Conics3_b` and `cov_Conics3_parabola_unique` in `Dandelin.Classical`; `cov_Conics4_a`, `cov_Conics5` and `cov_Conics5_double_root` in `Marden.Classical`.
- *Narrower* (1): for `classical_Conics4_b`, only the centre clause is proved (`Marden.Classical.cov_Conics4_b_part`: the inellipse is centrally symmetric about $(a+b+c)/3$). The area formula $\pi/(3\sqrt{3}) \cdot \text{area}(abc)$ and the strict maximality of the area among inscribed ellipses are missing. This paper does not claim them, and the libraries contain no measure theory.

- *Out of scope* (1): `classical_Conics3_c` is not proved. This is the planar converse: in an abstract plane, the locus $|XF| = e \cdot \mathrm{dist}(X, d)$ is an ellipse, parabola or hyperbola in standard form, with focus F and directrix d. Theorem 8 is a converse on the cone, not this statement. The comparer checked by hand that the statement as written is true.
- *Writer statement false as written*: none.

New proofs, which are not in the main files:

- `Dandelin.Classical.l15_tangent_cone_iff`: in $\mathbb{R}^3$, a sphere of positive radius meets every generator in exactly one point if and only if it is an axis sphere $(C_d, |d|s)$ with $d \neq 0$. Hence the uniqueness in Conics1_a, Conics2_a and Conics3_parabola_unique holds among all such spheres, not only among axis spheres.
- `Dandelin.Classical.l15_gen_segment`: the segment of every generator between the two contact circles has length $|d_1 - d_2| \cos \alpha$, which equals the focal constant.
- `Dandelin.Classical.l15_isGreatest`: the focal constant of the ellipse is the greatest distance between two of its points, that is, the major axis.
- The transverse axis, in `Dandelin.Classical.cov_Conics2_b`: the focal constant of the hyperbola is the least distance between its two branches.
- `Dandelin.Classical.l15_span_circle`, `Dandelin.Classical.l15_directrix_eq`: the affine span of a contact circle is the plane σ_d, and its intersection with the cutting plane is the directrix ℓ_d, which is a line when $m^2 < 1$ (equivalently, $\beta < \pi/2$).
- `Dandelin.Classical.l15_planeAxisAngle`: $\cos \beta = \sqrt{1 - m^2}$, so the paper's e is $\cos \beta / \cos \alpha$.
- `Marden.Classical.l15_strict_of_inter`: if a side line meets the ellipse curve only at the midpoint, then the strict tangency (ii) of Theorem 11 holds.
- `Marden.Classical.l15_marden_steiner`: the Marden ellipse lies in the closed triangle.
- `Marden.Classical.l15_equilateral_iff`: p' has a double root if and only if the triangle is equilateral.

The classical definitions and statements are reproduced verbatim in the Layer-15 files. The final gates list 158 declarations for Dandelin (85 published, 17 added by Layer 14, 56 by Layer 15) and 56 for Marden (18, 6 and 32). Both pass, and the tampering suites reject every planted defect.

8 Limitations and further work

- *Tangency.* Neither development uses a general notion of a tangent line or plane to a conic. For Dandelin, tangency of spheres and planes is expressed by distance and projection (Lemma 20). In the main files it is connected to Mathlib's `IsTangentAt` only in the normalised version (Remark 21). The Layer-14 restatements use `IsTangent` and `IsTangentAt` throughout (Section 7.1). For Marden, tangency is metric (Section 4.2). The equivalence with the differential notion is argued but not formalised.
- *Uniqueness among conics.* Theorem 15 is uniqueness among focal ellipses, not among all conics.

- *Not formalised.* We do not formalise the area of the Steiner inellipse ($\pi/(3\sqrt{3})$ times the area of the triangle) or its maximality among inscribed ellipses. We also do not formalise the planar converse that the locus $|XF| = e \cdot \operatorname{dist}(X, d)$ is a conic in standard form (Section 7.1).
- *Cones.* Only right circular cones are treated. The regime equivalences that need finite dimension and $\dim E \geq 3$ are proved under those hypotheses, and the two-line apex section only in $\dim E = 3$. For planes through the apex, only the double cone is treated; the one-nappe versions, which give rays, are not.
- *Reference geometry.* Mathlib has no ellipse, parabola or hyperbola, so the conics appear only as loci given by explicit equations. A Mathlib contribution could take the equation form as it stands, but would benefit from agreed definitions of these curves and bridges to them.

Acknowledgements and use of AI

The formal proofs, the numerical checking scripts and a draft of this paper were prepared with the help of large-language-model AI assistants. Every proof is checked by the Lean kernel, and the author takes responsibility for all content. The author compared each formal statement in Table 1 with its informal counterpart by hand. AI agents carried out the Layer-14 and Layer-15 checks of version 2 (Section 7.1). This includes collecting the classical statements from the literature, which no human has yet checked against the sources.

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